Daniel Hardesty Lewis

Recipient
Laura Kurgan & Kellen Funk
Columbia University
New York NY 10027
United States

May 15, 2024

Dear Professor Kurgan and Professor Funk,

I am writing to express my strong interest in the Graduate Research Assistant position working on the Cities of Bail project in collaboration with Columbia Law School. With my extensive background in geospatial analysis, data visualization, and web development, I believe I am uniquely qualified to make valuable contributions to this important project examining the impact of bail bonds on urban communities.

I could mention how I've headed up and completed one of the lead moonshot projects for a $40M disaster resiliency initiative in Texas, taking on multiple hats to wrangle the data and code on one of the largest supercomputers in the world and provide the first seamless topographical dataset produced for an entire state independently of the USGS, but you don't need that in this cover letter. **What you need is this:**

Key Strengths and Areas for Improvement

After reviewing your project, I have identified several strengths and critical areas for improvement to enhance efficiency, maintainability, and scalability.

Strengths:

- *Effective Use of Mapbox:* Excellent implementation of interactive maps using Mapbox GL JS.
- *Comprehensive Data Layers:* Robust integration of various demographic and economic data layers.

Critical Areas for Improvement:

- *Performance Optimization:* To improve performance, I recommend optimizing the GeoJSON files by simplifying geometries or utilizing vector tiles. This will ensure faster loading times and enhanced user experience.
- *Adopting a Modern Framework:* Migrating the codebase to a modern framework like React or Vue would enable faster development cycles, better state management, and a more maintainable component-based architecture. This approach will facilitate rapid prototyping and iterative improvements.
- *Separation of Data and Presentation:* Decoupling the data from the presentation layer by storing data configurations in separate files and loading them dynamically will greatly enhance the maintainability and scalability of the application.
- *Dynamic Layer Management:* Refactoring the code to use functions or loops for adding layers will reduce redundancy and improve code clarity. Implementing dynamic layer management will make the codebase more efficient and easier to extend.

Proposed Next Steps

- Adopt React or Vue and build UI components to dynamically load and manage data layers.
- Optimize data handling by refactoring loading and leveraging vector tiles for large datasets.
- Separate data configurations into files and load them dynamically.
- Standardize naming conventions and refactor code into reusable functions. Implement dynamic layer management.

The current approach provides a strong foundation, and by addressing these critical areas, you can significantly enhance the performance, scalability, and maintainability of your project. I would be excited to bring my technical skills and creative problem solving to help drive this project forward. The combination of geospatial analysis and critical examination of the justice system aligns closely with my own research interests.

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Please let me know if you would like any additional information. I look forward to the opportunity to discuss further how I could contribute to the Cities of Bail project and the mission of the Center for Spatial Research.

Thank you for your consideration.

If you are interested, feel free to reach out: (713) 371-7875.

Oh and if you are interested in that topographical data: summitgeospatial.com

Sincerely,

Daniel Hardesty Lewis

Attached: curriculum vitæ

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Education

2025 ant. **M.S., Urban Planning**, *Columbia University*, New York City, United States

2017 Dec. **B.S., Mathematics**, *University of Texas*, Austin, United States

2017 Dec. **Certificate, Scientific Computation**, *University of Texas*, Austin, United States

Experience

Professional

2023 Nov. – Present **Founder**, *Summit Geospatial*, Austin, United States
Detailed accomplishments:
- Led a team that developed web sales portal of the highest quality seamless terrain data in Texas

2021 Aug. – 2023 Aug. **Engineering scientist associate**, *Texas Advanced Computing Center*, Austin, United States
Detailed accomplishments:
- Principal developer of the highest quality seamless terrain data in Texas
- Lead of a principal project of a $40M disaster resiliency initiative
- Validated data from top State hydrologists and eminent UT engineers
- Developed the licensing & commercialization strategy for this data, secured approval of these strategy from director of IP without substantive changes
- Achieved a 14% click-through rate (CTR) with key audiences
- Helped secure multi-million dollar contracts with the State of Texas and the US Army Corps of Engineers
- Teach undergrads, grads, post-docs, professional researchers from UT Austin, University of Cambridge, & Texas A&M - Galveston

2018 Feb. – 2021 July **Research engineering / scientist associate I**, *Texas Advanced Computing Center*, Austin, United States
Detailed accomplishments:
- Contributed research in integrated modelling and water resource management in Texas and abroad
- Collaborated to aggregate data of Texas's water resources for integrated hydrology modellers

Teaching

2020 Aug. – 2020 Dec. **Co-instructor**, *University of Texas*, Austin, United States
Scientific computation Consulted with students for their homework and class projects

2018 June – 2018 Aug. **Teaching assistant**, *Texas Advanced Computing Center*, Austin, United States
Machine learning for the geosciences Recommended high performance optimisations for Petrobras's code

2018 Feb. – 2018 May **Co-instructor**, *University of Texas*, Austin, United States
Data informatics and intelligent systems for the geosciences Designed and instructed the technical half of this graduate course: intermediate programming techniques, Linux, and introductory high-performance computing

Peer-reviewed articles

2021 July Gil, Y., D. Garijo, D. Khider, C. A. Knoblock, M. Osorio, H. Vargas, M. Pham, J. Pujara, B. Shbita, B. Vu, Y. Chiang, D. Feldman, Y. Lin, H. Song, V. Kumar, A. Khandelwal, M. Steinbach, K. Tayal, S. Xu, S. A. Pierce, L. Pearson, *D. Hardesty Lewis*, E. Deelman, R. F. da Silva, R. Mayani, A. R. Kemanian, Y. Shi, L. Leonard, S. .D. Peckham, M. Stoica, K. M. Cobourn, Z. Zhang, C. Duffy, L. Shu. "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". The ACM Transactions on Interactive Intelligent Systems 11(2):11.

Technology skills

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Languages	Python, Bash shell, Fortran, Perl	GIS	ArcGIS, QGIS, PostGIS, GeoServer, GeoNode
Virtualization	Docker	Databases	PostgreSQL, SQL
Scientific models	HAND, HEC-RAS	Libraries	GDAL/OGR, GEOS, NumPy, GeoPandas, RasterIO, Matplotlib, Plotly, Dash, Vue, D3, React, Vue, Mapbox, GeoJSON
Cloud	AWS, Azure, GCP	Source code management	Git

Review committees

2023 Mar. – 2023 Apr.
Frontera Fellows applications
Determined the most qualified doctoral fellows' projects on one of the largest supercomputers in the world

2022 Aug.
TACC's Frontera Pathways & Large-Scale Community Partnerships applications
Reviewer approving/denying millions of dollars in computing resources to multimillion-dollar projects

Students mentored

2021 Aug. – Present
Khushboo Agarwal
MS Energy and Earth Resources, the University of Texas at Austin (expected 2023)

2020 Aug. – Present
Mark Wang
PhD Civil Engineering, the University of Texas at Austin (expected 2025)
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2022)

2019 Aug. – 2021 May
Robert Schomp
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2021)

2019 Aug. – 2020 May
Alec Carruthers
MS Environmental Engineering Technology / Environmental Technology, the University of Texas at Austin (2020)

Academic projects

2024 Apr.–Present
Massive Tracking
Developed the most comprehensive open-source call data record to mobility metric. *Preparing results for publication.*

2024 Apr.–Present
Dallas Mortgage Analysis
Implemented scraper for parcel-level, de-anonymized mortgage data for Dallas, Texas. This data tracks closely with actual sales data, previously unavailable at the individual home level in Texas, and is more accurate than Zillow. *Seeking research collaborators for this project.*

2024 Apr.–Present
Geography of NIMBYism in Texas
Scoping out costs and procedures to obtain protest letter data opposing zoning changes in major Texas cities via FOIA requests. Data includes addresses of non-government protestors within 200 ft of proposed changes. *Seeking research collaborators for this project.*

2022 June – 2023 Aug.
Advanced compound flood hazard assessment science
Partner with US Army Corps of Engineers to statistically model flood hazards in Texas's coastal region

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2021 Oct. – 2023 Aug. Texas Disaster Information System (TDIS)
Provide analytical web-based spatial data system to support more resilient decision-making at the state level

2020 Dec. – 2023 Aug. Museum of South Texas History Sunday Speaker Series
Provide Dash-enabled website to facilitate geolocation of archival imagery by museum visitors

2020 Nov. – 2023 Aug. Networks for Hazard Preparedness and Response
Overlay new, finely tuned flood maps over specially designed building and social vulnerability maps

2020 Sep. – 2023 Aug. Real-time flood inundation mapping to improve community resilience
Provide computationally efficient, high-resolution flood inundation maps to emergency response personnel

2020 Jan. – 2023 Aug. Estimating Inundation Extent and Depth from National Water Model Outputs and High Resolution Topographic Data
Improving the accuracy of flood inundation estimation products 100-fold, from 10m to 1m resolution

Languages

French	Professional working	Spanish	Native

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